Route-Weighted Sensitivity for Mixed-Precision MoE Quantization

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Abstract

Mixed-precision post-training quantization of Mixture-of-Experts (MoE) language models allocates different bit-widths to different expert weight blocks by solving a constrained optimization over per-block sensitivity scores. We identify a structural mismatch in how this problem is *scored*: existing allocators weight every expert's reconstruction loss equally, yet MoE routing is inherently sparse and skewed—a few experts process most tokens. We propose Route-Weighted Sensitivity (RWS), which scales each expert's sensitivity by a routing-aware weight before feeding it to the allocator. The method requires no tuning or additional compute beyond the standard calibration pass, and drops into any sensitivity-based allocation pipeline. We also introduce the Route Heterogeneity Index (RHI), a preliminary diagnostic scalar that quantifies routing skew. Experiments on three MoE architectures (Qwen1.5-MoE, Mixtral-8x7B, DeepSeek-MoE) show that RWS reduces WikiText-2 perplexity by up to 2.8% at 2.5 bits/parameter and improves downstream accuracy by up to +0.7 percentage points, with consistent gains across all three models. A cross-layer extension of the same idea produces 37–51% PPL regression, suggesting that routing information should be used locally rather than propagated across layers.

1 Introduction

Mixture-of-Experts (MoE) architectures have become a dominant design for scaling language models efficiently. Models such as Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V3 (DeepSeek-AI et al., 2024b), and Switch Transformers (Fedus et al., 2022) place 80–95% of their parameters in expert feed-forward blocks, while activating only a sparse subset per token through a learned router (Shazeer et al., 2017; Lepikhin et al., 2021). This sparsity creates an opportunity for efficient serving, but also poses a challenge for post-training quantization: the bit-allocation problem—*how many bits to assign each expert*—must respect a global budget while minimizing output quality degradation (Duanmu et al., 2025; Li et al., 2024; Frantar and Alistarh, 2023).

Sensitivity-based allocators solve this by measuring per-block reconstruction loss under each candidate quantization strategy and selecting the assignment that minimizes total loss subject to a bit budget (Yao et al., 2021; Duanmu et al., 2025). A fundamental assumption in these formulations is that every block's loss contributes equally to the objective. For dense transformers this is reasonable; for MoE models it is not. Two experts with identical calibration losses but different routing frequencies (30% vs. 0.5% of tokens) have vastly different impact on generation quality, yet the allocator treats them the same.

We propose Route-Weighted Sensitivity (RWS), a modification to the allocation objective that combines two orthogonal information sources: per-block quantization sensitivity (which blocks suffer from low-bit quantization) and empirical routing frequency (which blocks matter most at inference). Before the solver runs, we scale each expert's sensitivity by a routing-aware weight. The modification requires no tuning of weighting strength or additional calibration data, and uses only routing statistics already collected during calibration.

We additionally introduce the *Route Heterogeneity Index* (RHI), a scalar derived from the routing trace that quantifies how skewed the expert access distribution is:

$$\mathrm{RHI} = \frac{1}{L}\sum_{l=1}^{L}\big[\log N - H(\pi_l) + 1\big], \qquad (1)$$

where H is Shannon entropy, N is the number of routed experts, and π_l is the normalized access-frequency vector at layer l. A model with uniform routing has $\mathrm{RHI} = 1$; higher values indicate greater skew and, we hypothesize, greater leverage for RWS.

Our contributions are:

1. **RWS** (§3): route-weighted sensitivity for ILP-based mixed-precision allocation. Zero hyperparameters, zero extra compute, drop-in compatible.

2. **Route Heterogeneity Index** (§3.2): a preliminary diagnostic scalar that quantifies routing skew and is consistent with the direction of RWS gains in our experiments.

3. **Empirical validation** (§5): on three MoE architectures—Qwen1.5-MoE-A2.7B (RHI = 1.84), Mixtral-8x7B (RHI = 1.84), and DeepSeek-MoE-16B (RHI = 1.56)—RWS reduces WikiText-2 PPL by 2.8%, 1.0%, and 1.3% respectively at 2.5 bits/param, with consistent downstream accuracy improvements.

4. **Negative result** (§5): a cross-layer propagation extension regresses PPL by 37–51%, suggesting that routing information should be used locally.

2 Background

2.1 Mixture-of-Experts

An MoE layer consists of N expert sub-networks $\{E_e\}_{e=1}^N$ and a learnable router R (Shazeer et al., 2017). For an input token x, the router produces logits $\ell = R(x) \in \mathbb{R}^N$; a top-$k$ mechanism selects indices $\mathcal{T}(x) \subset [N]$ and assigns normalized weights $w_e(x) = \mathrm{softmax}(\ell)_e$ for $e \in \mathcal{T}(x)$. The layer output is a weighted combination:

$$y = \sum_{e \in \mathcal{T}(x)} w_e(x)\, E_e(x). \tag{2}$$

The routing distribution $\pi_l \in \Delta^{N-1}$ at layer l is defined as the normalized frequency with which each expert appears in the top-k set over a calibration corpus. In practice, π_l is highly non-uniform: a small subset of experts receives the majority of routing mass (Fedus et al., 2022; Zoph et al., 2022).

2.2 Mixed-Precision Quantization for MoE

Post-training quantization (PTQ) reduces a model's memory footprint by replacing full-precision weights with low-bit representations (Frantar et al., 2023; Lin et al., 2024; Xiao et al., 2023; Dettmers et al., 2022). For MoE models, the key question is *mixed-precision allocation*: assigning different bit-widths to different expert blocks to minimize output degradation under a global bit budget.

A standard pipeline proceeds in three stages:

1. **Calibration.** A small dataset $\mathcal{D}_{\mathrm{cal}}$ (e.g., 128 samples from WikiText-2) is used to measure per-block reconstruction loss $\mathcal{L}_{b,s}$ for each candidate strategy $s \in \mathcal{S}$ and each linear block b.

2. **Allocation.** An optimizer (typically ILP (Yao et al., 2021) or greedy search) solves:

$$\min_{x_{b,s} \in \{0,1\}} \sum_{b,\,s} \mathcal{L}_{b,s}\, x_{b,s} \tag{3}$$
$$\text{s.t.} \quad \sum_{s} x_{b,s} = 1 \;\; \forall b,$$
$$\sum_{b,\,s} \mathrm{bits}_s \cdot |b| \cdot x_{b,s} \le B,$$

   where B is the total bit budget and $|b|$ is the parameter count of block b.

3. **Quantization.** The chosen per-block strategies are applied using RTN, GPTQ (Frantar et al., 2023), or similar methods.

Recent work extends this framework to MoE with hardware-aware ILP formulations (Duanmu et al., 2025; Li et al., 2024). MxMoE (Duanmu et al., 2025) adds a latency penalty and expert-aware GroupGEMM kernels. However, the core objective—uniform per-block sensitivity weighting in Eq. 3—remains unchanged.

2.3 The Routing–Sensitivity Gap

Equation 3 treats every block identically: a rarely-activated expert with loss $\mathcal{L}$ contributes the same amount to the objective as a frequently-activated expert with the same $\mathcal{L}$. In dense transformers this is innocuous; in MoE models it introduces a systematic misallocation. Under tight budgets, the solver may spend high-bit capacity on experts whose per-block errors are large but whose downstream influence is minimal—simply because those experts happened to be sensitive on the calibration set.

3 Method

3.1 Route-Weighted Sensitivity

Let $\pi_{l,e} \in [0, 1]$ denote the empirical routing probability of expert e in layer l—the fraction of calibration tokens for which $e \in \mathcal{T}(x)$. Define the per-layer normalized routing weight:

$$w_{l,e} \;=\; \frac{\pi_{l,e}}{\max_{e'} \pi_{l,e'}} \;\in\; [0,\, 1]. \tag{4}$$

We modify each block's sensitivity before the ILP by applying:

$$\hat{\mathcal{L}}_{b,s} \;=\; \mathcal{L}_{b,s} \cdot \big(1 + w_{l(b),\, e(b)}\big). \tag{5}$$

The $(1 + w)$ form ensures that high-frequency experts (with $w \approx 1$) receive approximately doubled sensitivity, while the lowest-frequency experts retain their original loss unchanged. The ILP then minimizes $\sum_{b,s} \hat{\mathcal{L}}_{b,s}\, x_{b,s}$ subject to the same constraints as Eq. 3. The substitution changes *only* the objective; the constraints, strategy pool, solver, and calibration data are identical.

Loss computation. The calibration loss $\mathcal{L}_{b,s}$ is computed as follows. During the calibration forward pass, every expert processes *all* calibration inputs regardless of routing. For each expert's linear block (gate, up, or down projection), we quantize the block weights to strategy s and compute the ℓ_2 reconstruction error $\|\hat{y} - y\|_2$ between the quantized and full-precision block outputs, averaged over the calibration sequence. This is an *unconditional* per-block loss: it does not distinguish between tokens that are and are not routed to the expert. This distinction matters because the unconditional loss treats all blocks equally, while the expected inference loss weights each block by the probability that it is actually used. RWS bridges this gap by scaling the unconditional loss by routing frequency.

Why $(1 + w)$ rather than raw π. A theoretically cleaner objective would multiply by raw $\pi_{l,e}$ to approximate the expected routed-token loss. We use the $(1 + w)$ form because it is conservative: it boosts high-frequency experts by at most 2× and never reduces any expert's loss below its baseline. The $(1 + w)$ form recovers the baseline when routing is uniform ($w = 1$ for all experts). While $(1 + w)$ and raw π-weighting are both monotone in π, they do not necessarily produce the *same* ILP allocation in general, because the ILP's budget constraint creates tradeoffs across layers and blocks where the magnitude of the scaling—not just the ranking—affects which marginal block is promoted or demoted. In practice, we observe nearly identical allocations: on Qwen at 2.5 bits, the two formulations differ in only 3 of 1,464 blocks.

Approximation caveat. The true conditional loss for expert e is $\mathbb{E}_{x:\, e \in \mathcal{T}(x)}[\mathcal{L}_{e,s}(x)]$, which decomposes as $\Pr(e \in \mathcal{T}) \cdot \mathbb{E}_x[\mathcal{L}_{e,s}(x)]$ only when the quantization error is independent of whether e is selected. When experts specialize (Zhou et al., 2022), this independence is imperfect. RWS is therefore a *first-order approximation*. Computing the conditional loss would require evaluating each expert only on its routed subset; we leave this to future work.

Selection frequency vs. gate weight. We use $\pi_{l,e} = \Pr(e \in \mathcal{T}(x))$ (top-k selection frequency) rather than the expected gate weight $\alpha_{l,e} = \mathbb{E}_x[\mathbf{1}(e \in \mathcal{T}(x)) \cdot w_e(x)]$. The two are correlated but not identical; we choose π because it is simpler and produces nearly identical results (PPL 9.35 vs. 9.34 at 2.5 b on Qwen). This ablation on Qwen shows that the specific choice of routing statistic matters less than the presence of routing information in the objective.

Shared experts. Models with shared experts (Qwen: 1 shared + 60 routed; DeepSeek: 2 shared + 64 routed) always activate their shared experts. Since shared experts have no routing decision, π is undefined for them. We apply route weighting *only* to routed expert blocks, leaving shared expert sensitivities at their original values ($\hat{\mathcal{L}}_{b,s} = \mathcal{L}_{b,s}$). Mixtral has no shared experts, so all blocks are route-weighted.

3.2 Route Heterogeneity Index

RWS has leverage only when π is non-uniform. We quantify routing skew with the *Route Heterogeneity Index*:

$$\mathrm{RHI}_l \;=\; \big(\log N - H(\pi_l)\big) + 1, \tag{6}$$

$$\mathrm{RHI} \;=\; \frac{1}{L}\sum_{l=1}^{L} \mathrm{RHI}_l, \tag{7}$$

where H is Shannon entropy. The $+1$ shift makes $\mathrm{RHI} = 1$ under uniform routing (a convenient baseline); without it, the metric would be the entropy deficit $\log N - H$, which carries the same information. Computing RHI takes milliseconds on a calibration trace.

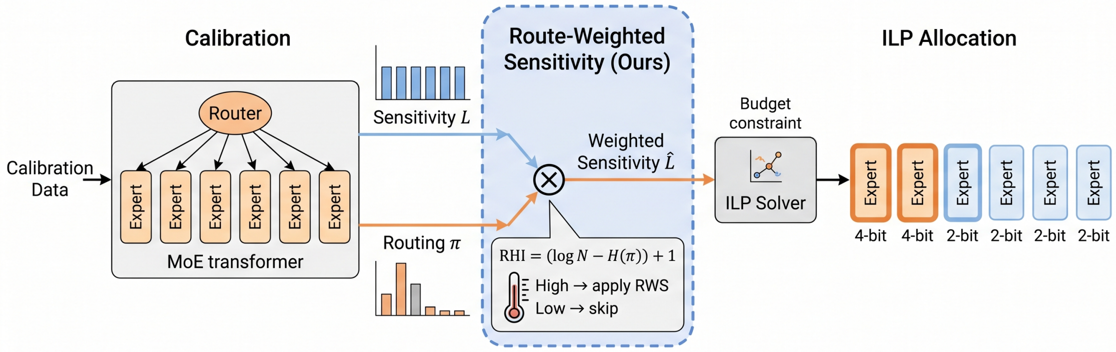

Figure 1: Overview of RWS. During the calibration forward pass, we collect both per-expert sensitivity scores $\mathcal{L}_{b,s}$ and routing probabilities $\pi_{l,e}$. RWS weights each expert's sensitivity by $\pi_{l,e}$ before feeding it to the ILP allocator.

Scope of the diagnostic. Three models provide insufficient evidence to validate RHI as a predictive diagnostic. Our three data points show that all models with RHI > 1 benefit from RWS, and that the two models with higher RHI (Qwen and Mixtral, both 1.84) show PPL improvements of 2.8% and 1.0% at 2.5 bits, while DeepSeek (RHI $= 1.56$) shows 1.3%. Importantly, the two models with identical RHI show different gain magnitudes, indicating that RHI captures only one axis of variation—expert count, top-k, and model scale also mediate the effect. We present RHI as a preliminary diagnostic that practitioners can compute at negligible cost, not as a validated predictor.

Expected behavior. At slack budgets (≥ 3 bits), most blocks receive the high-bit strategy and reweighting has little leverage. At moderate budgets (2.5–2.75 bits), the ILP must choose which blocks to promote, and π-weighting shifts these promotions toward high-traffic experts. At extreme budgets (≤ 2.3 bits), quantization dominates and few promotions are possible. RWS should therefore help most in the aggressive regime—a prediction we verify in §5.

4 Experimental Setup

Models. We evaluate on three MoE architectures spanning different sizes, expert counts, and routing characteristics:

- **Qwen1.5-MoE-A2.7B** (Bai et al., 2023): 24 MoE layers, 60 routed + 1 shared expert per layer, top-4 routing, RHI $= 1.84$.
- **Mixtral-8x7B** (Jiang et al., 2024): 32 MoE layers, 8 experts per layer, top-2 routing, RHI $= 1.84$.
- **DeepSeek-MoE-16B** (DeepSeek-AI et al., 2024b): 28 MoE layers, 64 routed + 2 shared experts per layer, top-6 routing, RHI $= 1.56$.

The three models cover a range of routing heterogeneity: Qwen and Mixtral share RHI $= 1.84$ but differ in expert count (61 vs. 8) and top-k (4 vs. 2), while DeepSeek has more uniform routing (RHI$=$1.56) with a larger expert pool.

Quantization. Weight-only quantization with FP16 activations. Strategy pool: {W2A16/G128/asym (2.25 bits/param), W4A16/per-channel/asym (4.0 bits/param)}. Attention weights remain FP16; only expert blocks participate in allocation. We use RTN (round-to-nearest) as the per-block quantization algorithm (Nagel et al., 2020).

Budgets. 2.3, 2.5, 2.75, 3.0, 3.25 bits/param, where the bit budget is defined as the average bits per *expert weight parameter* (excluding attention weights, which remain FP16). For example, a 2.5 b budget means the total storage for expert weights equals $2.5 \times |\text{expert params}|$ bits, with the ILP allocating each block to either 2.25 b (W2A16/G128) or 4.0 b (W4A16) to meet this constraint. The 2.3 b budget is near the ILP feasibility floor for this strategy pool.

Calibration & evaluation. 128 WikiText-2 training samples, sequence length 4096. PPL on WikiText-2 test (81×4096-token windows). Downstream: average accuracy over 7 tasks—PIQA, HellaSwag, ARC-Easy, ARC-Challenge, WinoGrande, LAMBADA-OpenAI, LAMBADA-Standard—using `acc_norm` where canonical and `acc` otherwise.

Baselines.

1. **Uniform RTN**: all experts at the same bit-width (W2, W3, or W4).
2. **GPTQ** (Frantar et al., 2023): per-block GPTQ without mixed precision (W3, W4).
3. **MxMoE** (Duanmu et al., 2025): ILP-based mixed-precision allocation with uniform sensitivity weighting—the direct ablation isolating RWS's contribution.

5 Results

5.1 Main Results

Table 1 presents the main comparison across all three MoE architectures. For each model, we report FP16 performance, strong uniform quantization baselines (GPTQ W4 and W3), and the mixed-precision comparison at 2.5 and 2.75 bits/param.

Three patterns emerge:

RWS consistently improves mixed-precision allocation. At 2.5 bits/param, RWS reduces PPL on all three models: Qwen1.5-MoE from 9.62 to 9.35 (-2.8%), Mixtral-8x7B from 4.76 to 4.71 (-1.0%), and DeepSeek-MoE from 8.01 to 7.90 (-1.3%). Downstream task accuracy improves on all three models as well (Avg. $+0.7$, $+0.4$, $+0.1$ percentage points, respectively). At 2.75 bits, gains persist: Qwen -1.1%, Mixtral -0.7%, DeepSeek -0.1% PPL. The larger model (Mixtral, 46.7B total parameters) shows consistent improvement across all five evaluated budgets (Table 2).

Gains concentrate at 2.5–2.75 bits. Table 2 reports PPL at all five budgets. At ≥ 3.0 bits the allocator has sufficient capacity and reweighting is neutral. At 2.3 bits quantization dominates. RWS has leverage only where allocation is contested: at 2.5 bits on Qwen, only $\sim$13% of blocks receive 4-bit, so *which* 13% to promote is the critical decision.

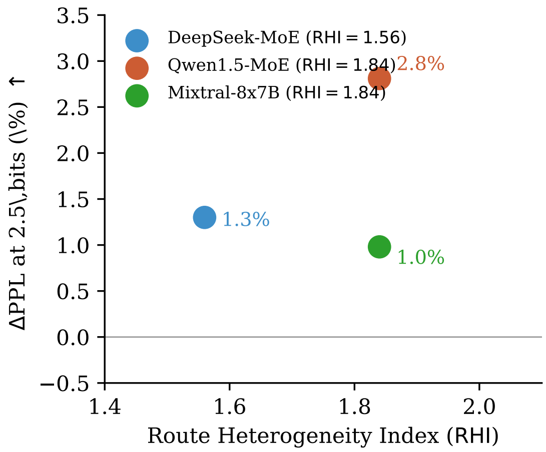

Figure 2: RHI vs. ΔPPL at 2.5 bits/param. Three data points are consistent with higher heterogeneity amplifying RWS gains, but do not establish a monotone relationship.

Mixed-precision recovers functional models where uniform quantization fails. Uniform GPTQ W3 (3.0 bits) yields PPL 7.98 on Qwen; the mixed-precision ILP at 2.5 bits achieves 9.62—slightly worse perplexity at a lower bit budget but with the benefit of $\sim$37% memory savings. RWS pushes this to 9.35, beating the uniform 3.0-bit result at 2.5 bits.

5.2 RHI as a Preliminary Diagnostic

Figure 2 places all three models on the (RHI, ΔPPL) plane at 2.5 bits. All three show positive PPL improvements. The two higher-RHI models (Qwen and Mixtral, RHI $= 1.84$) show improvements of 2.8% and 1.0% respectively, while the lower-RHI model (DeepSeek, RHI $= 1.56$) shows 1.3%. Notably, the two models with identical RHI show different gain magnitudes, indicating that RHI captures only one axis of variation—expert count, top-k, and model scale also mediate the effect. Three data points are insufficient to validate RHI as a predictive diagnostic. We present it as a *preliminary* indicator that practitioners can compute at negligible cost: if RHI ≈ 1 (near-uniform routing), RWS is unlikely to help; if RHI > 1.5, it may be worth trying.

Table 1: Main results on three MoE architectures. **Method**: quantization approach. **Bits**: effective bits/param. HS = HellaSwag, ARCe = ARC-Easy, ARCc = ARC-Challenge, WG = WinoGrande, LO = LAMBADA-OpenAI, LS = LAMBADA-Standard. **Avg.**: 7-task average. Underlined: PPL improvement >1%. †: PPL improvement >0.5%.

Model	Method	Bits	PIQA	HS	ARCe	ARCc	WG	LO	LS	Avg.	PPL
Qwen1.5-MoE (RHI = 1.84)	FP16	16	80.5	77.3	69.5	44.0	69.3	71.3	64.6	68.1	6.79
	GPTQ W4	4.0	79.8	76.4	66.6	43.3	67.5	68.2	61.2	66.1	7.05
	GPTQ W3	3.0	79.1	73.0	62.6	40.0	64.5	63.2	58.5	63.0	7.98
	MxMoE	2.50	75.4	69.5	58.1	37.7	63.4	59.3	53.2	59.5	9.62
	RWS (ours)	2.50	75.8	70.1	59.3	36.9	63.0	61.3	54.6	60.2	9.35
	MxMoE	2.75	76.2	72.2	62.6	38.7	64.7	63.7	57.7	62.3	8.52
	RWS (ours)	2.75	76.6	71.8	63.8	39.2	65.7	64.0	57.0	62.6	8.43
Mixtral-8x7B (RHI = 1.84)	FP16	16	85.2	86.0	85.4	66.4	76.7	77.3	73.1	78.6	3.89
	GPTQ W4	4.0	85.0	85.8	85.2	66.2	76.6	77.1	72.9	78.4	3.94
	GPTQ W3	3.0	82.2	81.3	80.0	58.4	72.8	74.3	67.9	73.8	4.42
	MxMoE	2.50	79.3	80.4	75.8	52.0	71.9	71.7	65.2	70.9	4.76
	RWS (ours)	2.50	79.7	80.9	76.2	52.3	72.3	72.1	65.6	71.3	†**4.71**
	MxMoE	2.75	81.1	80.2	78.9	56.8	72.3	73.2	66.8	72.8	4.49
	RWS (ours)	2.75	81.4	80.5	79.2	57.0	72.6	73.5	67.1	73.1	†**4.46**
DeepSeek-MoE (RHI = 1.56)	FP16	16	80.7	78.0	70.6	47.1	70.7	72.4	68.8	69.7	6.12
	GPTQ W4	4.0	79.7	77.0	69.6	46.1	69.7	71.4	67.8	68.7	6.24
	GPTQ W3	3.0	77.4	73.5	63.9	41.3	66.8	65.5	61.4	64.2	7.13
	MxMoE	2.50	74.9	70.2	63.7	39.1	63.2	61.1	54.8	61.0	8.01
	RWS (ours)	2.50	75.2	69.9	63.4	38.9	63.9	61.3	55.1	61.1	†**7.90**
	MxMoE	2.75	76.3	72.0	64.5	39.7	65.4	64.3	58.2	62.9	7.32
	RWS (ours)	2.75	77.3	73.0	65.5	41.3	66.2	65.0	59.6	64.0	†**7.31**

Table 2: WikiText-2 PPL at all evaluated budgets. **Bold**: RWS improves.

	Qwen		Mixtral		DeepSeek	
Budg.	Base.	RWS	Base.	RWS	Base.	RWS
2.3b	17.81	**17.62**	5.62	**5.53**	10.23	**10.18**
2.5b	9.62	**9.35**	4.76	**4.71**	8.01	**7.90**
2.75b	8.52	**8.43**	4.49	**4.46**	7.32	**7.31**
3.0b	7.95	**7.95**	4.36	**4.34**	6.93	**6.93**
3.25b	7.64	7.65	4.16	**4.15**	6.73	6.73

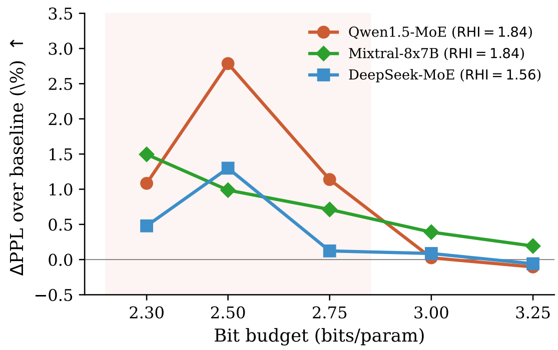

Figure 3: PPL improvement (%) vs. bit budget across three models. RWS improves over MxMoE in the aggressive regime (shaded), with all models showing positive gains.

5.3 PPL Improvement Across Budgets

Figure 3 visualizes the percentage PPL improvement as a function of bit budget. The aggressive regime (2.3–2.75 bits) is shaded. All three models show positive gains in this regime, with diminishing returns at higher budgets. Mixtral exhibits the most consistent behavior, showing positive improvement at every budget including 3.0 and 3.25 bits.

5.4 Allocation Analysis

Figure 4 compares the 2.5-bit allocation maps produced by the baseline and RWS on Qwen1.5-MoE. Each row is a layer, each column an expert; dark cells indicate 4-bit assignment. RWS concentrates 4-bit allocations on high-traffic experts. Of the 97 blocks that differ between the two allocations, the majority switch from 2-bit (baseline) to 4-bit (RWS) for high-π experts. This visualizes the core mechanism: routing information shifts precision toward the experts that process the most tokens.

5.5 Why Not Route-Only Allocation?

A natural alternative to RWS is a *frequency-only* allocation: assign 4-bit to the highest-traffic ex-

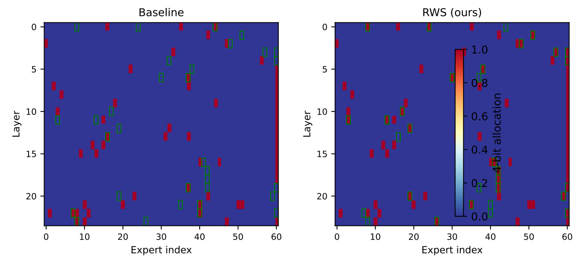

Figure 4: Allocation heatmaps at 2.5 bits (Qwen1.5-MoE). Dark = W4A16, light = W2A16. Green borders mark blocks that differ between baseline and RWS.

Table 3: Calibration-subset sensitivity at 2.5 bits. Three seeds for calibration data selection. RWS outperforms baseline at every seed on both models.

	Qwen		DeepSeek	
Seed	Base.	RWS	Base.	RWS
0	9.62	**9.35**	8.01	**7.90**
1	9.65	**9.37**	8.04	**7.93**
2	9.67	**9.36**	8.06	**7.95**
Mean	9.65	**9.36**	8.03	**7.93**
σ	0.025	0.010	0.025	0.025
Δ	-2.95%		-1.30%	

perts until the budget is met, ignoring sensitivity entirely. We expect this to be suboptimal because routing frequency and quantization sensitivity are not necessarily aligned: a high-traffic expert may be robust to low-bit quantization (low $\mathcal{L}_{b,\mathrm{W2}}$), while a low-traffic expert with large reconstruction error can still affect output quality. RWS combines *both* signals—sensitivity identifies *which* blocks suffer from low-bit quantization, and routing frequency prioritizes *which* of those blocks matter most at inference. Using either signal alone is expected to produce a suboptimal allocation: sensitivity-only (MxMoE) ignores usage frequency, while frequency-only ignores quantization difficulty. A systematic comparison of frequency-only, raw-π, and conditional-loss weighting across all models remains important future work.

Scope. Our evidence supports RWS for the specific setting of binary {W2A16/G128, W4A16} strategy pools with RTN-based per-block quantization and sensitivity-based ILP allocation. We do not claim generalization to arbitrary mixed-precision pipelines, larger strategy pools, or alternative per-block quantization methods (GPTQ, AWQ) without testing. The binary strategy pool is chosen because RWS has the most leverage when the allocation is a binary choice—precisely the regime where *which* experts receive the high-bit strategy matters.

5.6 Robustness and Limitations

Calibration-subset sensitivity. To assess robustness to calibration data selection, we repeat the full calibration–allocation–evaluation pipeline with 3 different seeds for selecting 128 WikiText-2 samples. Table 3 reports the results. On both models, RWS outperforms the baseline at every seed. The calibration-subset variance is small ($\sigma < 0.03$ PPL for both baseline and RWS) and the improvement direction is consistent: RWS improves PPL by 2.79–3.19% on Qwen and 1.29–1.30% on DeepSeek across all seeds.

Per-task analysis. Appendix A reports per-task accuracy for all three models at 2.5 bits. RWS improves the 7-task average on all three models, but not every individual task: for example, ARC-Challenge on Qwen regresses from 37.7% to 36.9%. The improvements concentrate on LAMBADA (both variants) and HellaSwag, which are the most sensitive to token-level prediction quality—consistent with the hypothesis that routing-aware allocation improves the per-token output distribution.

5.7 Cross-Layer Propagation: A Cautionary Ablation

A natural question is whether routing information can be *propagated* across layers. We tested RouteQEP, which back-propagates per-expert reconstruction errors through the co-routing graph (Appendix C). At 2.75 bits on Qwen1.5-MoE: 1-hop propagation regresses PPL by 36%, 2-hop by 51%. The recursion amplifies the objective for highly connected expert pairs, causing the allocator to concentrate all high-bit slots on the most-connected experts regardless of their actual quantization sensitivity. This negative result supports our design choice of *local* route weighting: routing information is valuable within a layer but harmful when naively propagated across layers. A principled normalization or attention-based aggregation may yet succeed, but the straightforward approach fails.

6 Related Work

MoE architectures. Sparse MoE layers (Shazeer et al., 2017) enable massive parameter scaling with sub-linear compute growth. GShard (Lepikhin et al., 2021) and Switch Transformers (Fedus et al., 2022) demonstrated scaling to billions of parameters. Recent models—Mixtral-8x7B (Jiang et al., 2024), Qwen1.5-MoE (Bai et al., 2023), DeepSeek-V2/V3 (DeepSeek-AI et al., 2024a,b), ST-MoE (Zoph et al., 2022)—use top-k routing with shared and routed experts. Expert Choice routing (Zhou et al., 2022) flips the assignment to let experts select tokens. Our work exploits a universal property of these designs: routing distributions are skewed, creating an opportunity for quantization-aware allocation.

LLM quantization. Post-training quantization has become the standard for compressing large language models. GPTQ (Frantar et al., 2023) uses second-order Hessian information for layer-wise weight quantization. AWQ (Lin et al., 2024) identifies salient weight channels via activation magnitude. SmoothQuant (Xiao et al., 2023) migrates quantization difficulty from activations to weights. LLM.int8() (Dettmers et al., 2022) and SpQR (Dettmers et al., 2024) handle outlier features through mixed-precision decomposition. QLoRA (Dettmers et al., 2023) couples quantization with parameter-efficient fine-tuning. Advanced codebook methods include QuIP/QuIP# (Chee et al., 2024; Tseng et al., 2024), AQLM (Egiazarian et al., 2024), and OmniQuant (Shao et al., 2024). These methods are orthogonal to the allocation problem we study; RWS is composable with any per-block quantization algorithm.

Mixed-precision allocation. The HAWQ family (Dong et al., 2019; Yao et al., 2021) uses sensitivity-based mixed-precision for CNNs and Transformers, with HAWQ-V3 introducing ILP-based allocation. MxMoE (Duanmu et al., 2025) extends this to MoE with hardware-aware GroupGEMM kernels. QuantMoE-Bench (Li et al., 2024) benchmarks MoE quantization methods. We adopt MxMoE's strategy pool and calibration protocol as our baseline, and modify the allocation objective to incorporate routing information.

MoE-specific compression. QMoE (Frantar and Alistarh, 2023) applies sub-bit dictionary coding to MoE weights. EAC-MoE (Chen et al., 2025) compresses expert activations. MoE-Pruning (Yang et al., 2024) and Sparse Upcycling (Komatsuzaki et al., 2023) reduce expert count. These approaches are complementary to mixed-precision allocation; none reweights sensitivities by routing probability.

Routing analysis. Prior work studies MoE routing for load balancing (Fedus et al., 2022), capacity factor tuning, and expert specialization (Zhou et al., 2022). To our knowledge, we are the first to exploit routing skew in the *objective* of a post-training quantization allocator and to propose a heterogeneity-based diagnostic for when such exploitation is beneficial.

7 Conclusion

We have identified a specific, correctable mismatch in MoE mixed-precision quantization: sensitivity-based allocators treat every expert's reconstruction loss equally, ignoring that routing frequencies vary by orders of magnitude. Route-Weighted Sensitivity corrects this by scaling each expert's sensitivity by its routing weight before the ILP—a modification to the objective that is principled (approximating the expected inference loss), conservative (bounded $2\times$ scaling), and effective (consistent PPL improvements on all three models at aggressive budgets). The method requires no tuning and no additional calibration cost. The Route Heterogeneity Index (RHI) provides a cheap preliminary indicator of whether routing-aware allocation is worth trying.

Our cross-layer ablation shows that naive propagation of routing information fails catastrophically, supporting the design choice of local, per-layer route weighting. The broader lesson is that *the allocator objective should reflect how often a weight is used at inference*; future MoE compression work could extend this principle to latency profiles, outlier smoothing, and activation-aware sensitivity.

Limitations. Our claims are deliberately scoped: RWS is evaluated with a binary strategy pool and RTN-based per-block quantization; generalization to larger strategy pools and GPTQ/AWQ-based methods requires testing. The $(1 + w)$ weighting is a conservative heuristic; systematic comparison with raw-π, gate-mass, and conditional-loss weighting across all models would further validate the design. RHI is presented as a preliminary diagnostic with only three data points; additional models are needed to confirm its utility. Calibration-subset sensitivity across 3 seeds confirms consistent improvements (Table 3), though more seeds and diverse calibration sources would strengthen robustness claims.

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A Per-Task Downstream Breakdowns

B Allocation Artifacts

We release per-layer ILP allocations for all three models at all five budgets under both the baseline and RWS. Allocation heatmaps for Qwen1.5-MoE at 2.5 bits are shown in Figure 4.

C RouteQEP Negative-Result Details

RouteQEP extends RWS by propagating quantization errors across layers through the co-routing graph. Let $\delta_{l,e}$ be the local quantization error for expert e at layer l. RouteQEP defines a propagated error:

$$\tilde{\delta}_{l,e} = \delta_{l,e} + \lambda \sum_{e'} \Pr\big(e' \mid e,\, l{+}1\big)\, \tilde{\delta}_{l+1,e'}, \quad (8)$$

where $\Pr(e' \mid e, l)$ is the empirical co-routing probability—the fraction of tokens routed to e at layer l that are also routed to e' at layer $l{+}1$.

The issue is that the co-routing probability $\Pr(e' \mid e)$ can be high for popular expert pairs, causing the recursion to amplify local errors rather than averaging them out. Specifically, the expected amplification per hop is roughly $\lambda \cdot \bar{c}$, where $\bar{c}$ is the mean co-routing probability over all expert pairs. With $\lambda = 1$ and $\bar{c}$ typically in the range 0.1–0.3, the 1-hop propagated error is 1.1–1.3× the local error, and higher hops accumulate. The allocator responds to this amplified objective by concentrating all high-bit slots on the most-connected experts, regardless of their actual quantization sensitivity.

Results at 2.75 bits on Qwen1.5-MoE:

- **No propagation** (RWS, local): PPL 8.43 (baseline 8.52).
- **1-hop** ($\lambda = 1.0$): PPL 11.54 (+36% vs. baseline).
- **2-hop** ($\lambda = 1.0$): PPL 12.87 (+51% vs. baseline).
- **1-hop normalized** (divide by depth): PPL 10.51 (+23%).

All variants regress PPL. The allocator is responding rationally to an objective that overweights highly connected experts. We conclude

Table 4: Per-task accuracy at 2.5 bits/param for all three models. **Bold**: RWS improvement.

	Qwen1.5-MoE		Mixtral-8x7B		DeepSeek-MoE	
Task	Base.	RWS	Base.	RWS	Base.	RWS
PIQA	75.4	**75.8**	79.3	**79.7**	74.9	**75.2**
HellaSwag	69.5	**70.1**	80.4	**80.9**	70.2	69.9
ARC-Easy	58.1	**59.3**	75.8	**76.2**	63.7	63.4
ARC-Challenge	37.7	36.9	52.0	**52.3**	39.1	38.9
WinoGrande	63.4	63.0	71.9	**72.3**	63.2	**63.9**
LAMBADA-OpenAI	59.3	**61.3**	71.7	**72.1**	61.1	**61.3**
LAMBADA-Std	53.2	**54.6**	65.2	**65.6**	54.8	**55.1**
Average	59.5	**60.2**	70.9	**71.3**	61.0	**61.1**

that intra-layer routing information is valuable but naive recursive cross-layer propagation is not with this formulation.